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| research | Main points | Description and scope | abstract | |
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| Health Behavior Change in HCI: Trends, Patterns, and Opportunities | • **Chronic Disease Prevention**: Digital technologies can support health behavior change to prevent chronic diseases.<br>• **Research Trends**: There is a stronger focus on physical activity than on other health behaviors.<br>• **Demographic Focus**: Most studies involve adults, with limited attention to children and the elderly.<br>• **Evaluation Standards**: A lack of standardized methods exists for evaluating user experiences in health interventions.<br>• **Research Opportunities**: The paper suggests six areas for future research, including game integration and AI applications. | The paper explores the intersection of health behavior change and Human-Computer Interaction (HCI), examining existing research trends and identifying gaps and opportunities for further study. It provides insights into how digital interventions can be better designed and evaluated to promote healthier lifestyles. | Unhealthy lifestyles could cause many chronic diseases, which bring patients and their families much burden. Research has shown the potential of digital technologies for supporting health behavior change to help us prevent these chronic diseases. The HCI community has contributed to the research on health behavior change for more than a decade. In this paper, we aim to explore the research trends and patterns of health behavior change in HCI. Our systematic review showed that physical activity drew much more attention than other behaviors. Most of the participants in the reviewed studies were adults, while children and the elderly were much less addressed. Also, we found there is a lack of standardized approaches to evaluating the user experience of interventions for health behavior change in HCI. Based on the reviewed studies, we provide suggestions and research opportunities on six topics, e.g., game integration, social support, and relevant AI application. | يشمل البحث مراجعة منهجية للأدبيات المتعلقة بتغيير سلوكيات الصحة في مجال التفاعل بين الإنسان والآلة، مع التركيز على الفجوات الديموغرافية مثل نقص الدراسات التي تستهدف الأطفال وكبار السن. كما يناقش كيفية استخدام التقنيات الرقمية والذكاء الاصطناعي في تصميم التدخلات الصحية، ويقدم توصيات لدراسات مستقبلية تهدف إلى تحسين تلك التدخلات من خلال استراتيجيات جديدة ومبتكرة. |
| MHealth: An Artificial | **Objective**: The study introduces an expert system-based mHealth application that leverages | The research focuses on the development of a | Main objective of this study is to | هدف هذه الدراسة إلى تقديم تطبيق موبايل قائم |

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| Intelligence Oriented Mobile Application for Personal Healthcare Support | Artificial Intelligence to enhance personal healthcare support.<br>**Design and Development**: A mobile software system was designed to provide dynamic support for common health problems encountered in daily life.<br>**Evaluation**: The application was assessed through surveys and diagnosis-based evaluation tasks, yielding positive results.<br>**Literature Integration**: The application considers previously introduced solutions and requirements from existing literature for improved healthcare support. | mobile application that utilizes Artificial Intelligence to provide personal healthcare support. It emphasizes offering effective solutions for common health problems encountered in daily life. The study also addresses the evaluation of the application's effectiveness through user interactions and feedback, suggesting future improvements based on the results of these evaluations. Additionally, it incorporates insights from existing literature to enhance the application's design and functionality, aiming for a comprehensive approach to personal health management. | introduce an expert system-based mHealth application that takes Artificial Intelligence support by considering previously introduced solutions from the literature and employing possible requirements for a better solution. Thanks to that research study, a mobile software system having Artificial Intelligence support and providing dynamic support against the common health problems in daily life was designed-developed and it was evaluated via survey and diagnosis-based evaluation tasks. Evaluation tasks indicated positive outcomes for the mHealth system | على نظام خبير يستفيد من الذكاء الاصطناعي لتعزيز الدعم الشخصي للرعاية الصحية. تم تصميم نظام البرمجيات المحمولة لتوفير دعم ديناميكي لمشاكل الصحة الشائعة في الحياة اليومية. تم تقييم التطبيق من خلال استبيانات ومهام تقييم قائمة على التشخيص، مما أسفر عن نتائج إيجابية. يأخذ التطبيق في اعتباره الحلول السابقة والمتطلبات من الأدبيات الموجودة لتحسين دعم الرعاية الصحية. |
| Applications of Artificial Intelligence and Big Data Analytics in m-Health: A Healthcare System Perspective | **Objective**: The systematic review explores how Artificial Intelligence (AI) and big data analytics are applied in mobile health (mHealth) systems.<br>**AI Algorithms**: It discusses various AI-based algorithms and frameworks, detailing their relevance to different data sources and techniques used in mHealth.<br>**Application Areas**: The review highlights specific areas where AI can enhance personalized health planning within mHealth applications.<br>**Insights**: The findings provide a comprehensive understanding of the benefits and potential of integrating AI and big data in healthcare management. | The research focuses on the integration of AI and big data analytics in mHealth systems, examining how these technologies can improve healthcare delivery and personalized health planning. It provides an overview of different AI algorithms and their applications, highlighting the techniques employed and the data sources utilized. The study aims to inform healthcare professionals and developers about the potential of AI to enhance the effectiveness and personalization of mHealth interventions. | Mobile health (m-health) is the term of monitoring the health using mobile phones and patient monitoring devices etc. It has been often deemed as the substantial breakthrough in technology in this modern era. Recently, artificial intelligence (AI) and big data analytics have been applied within the m-health for providing an effective healthcare system. Various types of data such as electronic health records (EHRs), medical images, and complicated text | تستعرض هذه المراجعة المنهجية تطبيق الذكاء الاصطناعي وتحليلات البيانات الضخمة في نظم الصحة المحمولة (mHealth). تناقش الورقة مجموعة متنوعة من الخوارزميات والإطارات المعتمدة على الذكاء الاصطناعي فيما يتعلق بمصادر البيانات والتقنيات المستخدمة ومجالات التطبيق، مما يوفر رؤى حول كيفية تعزيز الذكاء الاصطناعي للتخطيط الصحي الشخصي ضمن تطبيقات الصحة المحمولة. |

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| | | | which are diversified, poorly interpreted, and extensively unorganized have been used in the modern medical research. This is an important reason for the cause of various unorganized and unstructured datasets due to emergence of mobile applications along with the healthcare systems. In this paper, a systematic review is carried out on application of AI and the big data analytics to improve the m-health system. Various AI-based algorithms and frameworks of big data with respect to the source of data, techniques used, and the area of application are also discussed. This paper explores the applications of AI and big data analytics for providing insights to the users and enabling them to plan, using the resources especially for the specific challenges in m-health, and proposes a model based on the AI and big data analytics for m-health. Findings of this paper will guide the development of techniques using the combination of AI and the big data as source for handling m-health data more effectively. | |

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| https://pmc.ncbi.nlm.nih.gov/articles/PMC7077036/ | This study explores the effectiveness of personalized nutrition apps that provide tailored dietary recommendations based on user inputs and preferences. It evaluates the impact of these apps on user adherence to dietary guidelines. | Focuses on user engagement, satisfaction, and the effectiveness of personalized dietary advice in improving nutritional habits. | | |
| https://www.mdpi.com/2071-1050/12/3/911 | This research examines how mobile applications can promote healthy eating behaviors by providing meal planning, calorie tracking, and personalized dietary recommendations. | Investigates the features that enhance user engagement and the effectiveness of these apps in improving dietary habits. | | |
| https://pmc.ncbi.nlm.nih.gov/articles/PMC6342588/ | This systematic review analyzes various mobile apps designed for weight management, focusing on their functionalities, including dietary tracking and exercise recommendations. | Evaluates the impact of these apps on weight loss outcomes and user satisfaction. | | |
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